

4.8.1 Carbon chemistry

The study of forms of carbon provides an opportunity to revisit ideas about structure and bonding from [Chemical quantities](#) (page 89). Other ideas from [Structure and bonding](#) (page 98) are applied to explain how new chemicals and materials are made from the hydrocarbons in crude oil. Featured processes include fractional distillation, cracking and polymerisation.

4.8.1.1 Bonding and structure in forms of carbon

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the properties of diamond, graphite, fullerenes and graphene in terms of their structures and bonding.	Diamond is very hard, has a very high melting point and does not conduct electricity. In diamond, each carbon atom forms four covalent bonds with other carbon atoms in a giant covalent structure. Graphite is soft, has a high melting point and conducts electricity. In graphite, each carbon atom forms three covalent bonds with three other carbon atoms, forming layers of hexagonal rings. There are no covalent bonds between layers. One electron from each carbon is delocalised. Graphene is a single layer of graphite and so is one atom thick. It has properties that make it useful in electronics and composites. Fullerenes are molecules of carbon atoms with hollow shapes. The structure of fullerenes is based on hexagonal rings of carbon atoms but they may also contain rings with five or seven carbon atoms. The first fullerene to be discovered was buckminsterfullerene (C₆₀), which has a spherical shape. Carbon nanotubes are cylindrical fullerenes with very high length to diameter ratios. Their properties make them useful for nanotechnology, electronics and materials.	MS 5b Visualise and represent 2D and 3D forms including two-dimensional representations of 3D objects. WS 1.4 Give examples of the uses of diamond, graphite and fullerenes, including carbon nanotubes.

4.8.1.2 Hydrocarbons in crude oil

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Recall that crude oil is a main source of hydrocarbons and is a feedstock for the petrochemical industry. Recognise that crude oil is a finite resource. Recall that carbon can form four covalent bonds. Explain that the vast array of natural and synthetic organic compounds occur due to the ability of carbon to form families of similar compounds, chains and rings. Describe the fractions as largely a mixture of compounds of formula C_nH_{2n+2} which are members of the alkane homologous series.	Crude oil is a finite resource found in rocks. Crude oil is the remains of an ancient biomass consisting mainly of plankton that was buried in mud. Crude oil is a mixture of a very large number of compounds. Most of the compounds in crude oil are hydrocarbons, which are molecules made up of hydrogen and carbon atoms only. Alkane molecules can be represented in the following forms: C_2H_6 or <pre> H H H — C — C — H H H </pre> Knowledge of the names of specific alkanes other than methane, ethane, propane and butane is not required.	WS 1.2, MS 5b Recognise substances as alkanes given their formulae.

4.8.1.3 Fractional distillation of crude oil

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe and explain the separation of crude oil by fractional distillation. Describe, explain and exemplify the processes of fractional distillation. Explain how modern life is crucially dependent upon hydrocarbons.	Some properties of hydrocarbons depend on the size of their molecules, including: • boiling point • viscosity, and • flammability. These properties influence how hydrocarbons are separated and how they are used as fuels. Knowledge of trends in properties of hydrocarbons is limited to boiling point, viscosity and flammability. The many hydrocarbons in crude oil may be separated into fractions, each of which contains molecules with a similar number of carbon atoms, by fractional distillation. The fractions can be processed to produce fuels and feedstock for the petrochemical industry. Many of the fuels on which our modern lifestyle depends such as petrol, diesel oil, kerosene, heavy fuel oil and liquefied petroleum gases, are produced from crude oil. Knowledge of the names of other specific fractions or fuels is not required. The combustion of hydrocarbon fuels releases energy. During combustion, the carbon and hydrogen in the fuels are oxidised. The complete combustion of a hydrocarbon produces carbon dioxide and water. Many useful materials on which modern life depends are produced by the petrochemical industry. These include solvents, lubricants, polymers and detergents.	WS 1.2 Write balanced equations for the complete combustion of hydrocarbons with a given formula. Relate trends in the hydrocarbons to molecular size using ideas in Covalent bonding (page 101).

4.8.1.4 Cracking hydrocarbons

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the production of materials that are more useful by cracking.	Hydrocarbons can be broken down to produce smaller, more useful molecules by catalytic cracking or by steam cracking. The products of cracking include alkanes and another type of hydrocarbon called alkenes. Recall of the formulae or names of individual alkenes, other than ethene, is not required. There is a high demand for fuels with small molecules and so some of the products of cracking are useful as fuels. Alkenes are used to produce polymers and as starting materials for the production of many other chemicals. Small ethene molecules polymerise to produce long-chain molecules of poly(ethene) (see also Covalent bonding (page 101)).	WS 1.2 Balance chemical equations as examples of cracking given the formulae of the reactants and products.